



AY 2024/2025 Semester 2
DSA4263 – Sense-making Case Analysis: Business and Commerce
Final Group Report

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Table of Contents

1. Abstract	2
2. Introduction	2
3. Data	2
3.1 Labels	3
3.2 Collection & Potential Biases	3
3.3 Missing and Irregular Data Values	4
4. Methods	4
4.1 Feature Engineering	4
4.2 Missing Data in the Aggregated Dataset: Identification and Imputation	7
4.2.1 Identification	7
4.2.2 Imputation	8
4.3 EDA Insights	8
4.3.1 Insights	8
4.3.2 Hypotheses	10
4.4 Feature Selection	17
5. Graph Construction	18
5.1 Similarity Graph	18
5.2 Shared IMEI Graph	20
5.3 VOC and SMS Interactions Graph	21
6. Models	22
6.1 Baseline Model - Relational Neighbour Classifier (RNC)	22
6.2.1 Care-GNN model	22
7. Models	24
7.1 Training-Testing Approach and Chosen Metrics	24
7.2 Model Results	25
7.3 Hyperparameter Tuning via Grid Search	27
7.4 Explainability Tools On CARE-GNN	28
7.4.1 Relationship Importance	28
7.4.2.1 SHAP	30
7.4.2.2 Integrated Gradient	31
8. Discussion/Conclusion	34
8.1 Summary	34
8.2 Contents	35
8.3 Considerations	35
9. Appendix	37
9.1 Data Dictionary	37
9.2 Additional Graph Construction	42
9.3 All Model Results	44

1. Abstract

Telecommunication fraud remains a significant and evolving threat in China, characterized by sophisticated behaviors that often elude traditional detection techniques. This study investigates the application of the CAmouflage-REsistant Graph Neural Network (CARE-GNN), a curriculum-aware GNN model, for fraudulent phone numbers detection within a real-world telecom dataset. The dataset includes App Usage, SMS, Voice Call, and User Consumption records, represented in 3 relations graphs - similarity, shared IMEI, and SMS/Call interactions. Despite fraudsters comprising 32% of the network—higher than typical real-world proportions—the dataset still presents several challenges common in telecom fraud detection: noisy features and edges, scalability in large-scale networks, and class imbalance.

CARE-GNN effectively addresses feature camouflage, where fraudsters mimic legitimate user profiles, and relation camouflage, where they establish connections with benign users. By aggregating neighbor information per relation type and learning their importance, the model produces discriminative node embeddings for accurate classification. Compared to other models, including Relational Neighbor Classifier (RNC) and feature-level ensemble approach, CARE-GNN outperforms with the highest AUC (0.953), showcasing enhanced robustness and detection accuracy.

We recommend CARE-GNN for environments with adversarial behavior and heterogeneous interactions. To bolster future fraud prevention, we advocate for collaborative efforts among telecom providers to share blacklists and user histories securely, particularly for addressing cross-border scams.

2. Introduction

While mobile internet and digital payment systems offer convenience, they have also fueled a rise in telecom fraud. Between 2023 and 2024, China saw a 27% increase in telecom and online fraud cases, with over 40,000 incidents reported in 2024 (Lee, 2024) and losses reaching 4.84 billion USD in 2020 (Wang, 2021).

The most common forms of telecom fraud in China involve digital communication channels like phone calls, SMS, and messaging apps. These include job scams, phishing via SMS, lottery scams, and impersonation, where scammers pose as customer service agents from Taobao to trick victims into sharing personal information or transferring funds (Xinhua, 2024).

Sichuan Province, the focus of our dataset, is a hub for fraud networks due to its central location and cross-provincial links (Liu et al., 2024). A rural-urban digital literacy gap further increases vulnerability (Ye et al., 2023), leading to financial losses and eroded trust in telecom systems.

While recent research on Graph Neural Networks (GNNs) shows promise for fraud detection, challenges remain, such as noisy edges and features in inferred relationships, scalability issues in large-scale networks, and class imbalance (Liu et al., 2024). To address these, this study proposes scalable and adaptive graph-based fraud detection models tailored to Sichuan's

telecommunication data to identify suspicious behavior patterns and improve detection performance. The paper also outlines the dataset, preprocessing, graph construction, CARE-GNN and ensemble methods, evaluation, and real-world implications.

3. Data

The dataset, from the 2020 Digital Sichuan Innovation Competition by Sichuan Big Data Center, contains anonymized telecommunications data for 6,106 subscribers across 23 Sichuan cities from August 2019 to March 2020. It spans four subsets of subscriber activity:

Data Subset	Rows	Columns	Description	Features
App Usage (application usage data)	3,283,602	4	Application Usage Data	phone_no_m, busi_name, flow, month_id
SMS Records (short message details)	6,848,509	4	Short Message Details	phone_no_m, opposite__no_m, calltype_id, request_datetime
Voice Call Logs (call records)	5,015,430	8	Call Records	phone_no_m, opposite_no_m, calltype_id, start_datetime, call_dur, city_name, county_name, imei_m
User Consumption (monthly spending and consumption patterns)	6,106	14	Monthly Spending and Consumption Patterns	phone_no_m, city_name, county_name, idcard_cnt, arpu_201908, arpu_201909, arpu_201910, arpu_201911, arpu_201912, arpu_202001, arpu_202002, arpu_202003, arpu_202004, label

Fig 1: Description of the dataset from the 2020 Digital Sichuan Innovation Competition

3.1 Labels

The dataset uses binary labels: 1 for fraudulent numbers and 0 for non-fraudulent numbers. These will be referred to as fraud and non-fraud respectively from here on.

3.2 Collection & Potential Biases

Collected and anonymized by China Mobile Sichuan Corporation, this dataset, while comprehensive, may contain potential biases. Since it was used in a competition, the minority class might have been slightly oversampled to make trends more apparent. With fraudulent cases still making up 32.13% of the data, this percentage is likely higher than what would be observed in a real-world scenario (Hiya, 2024).

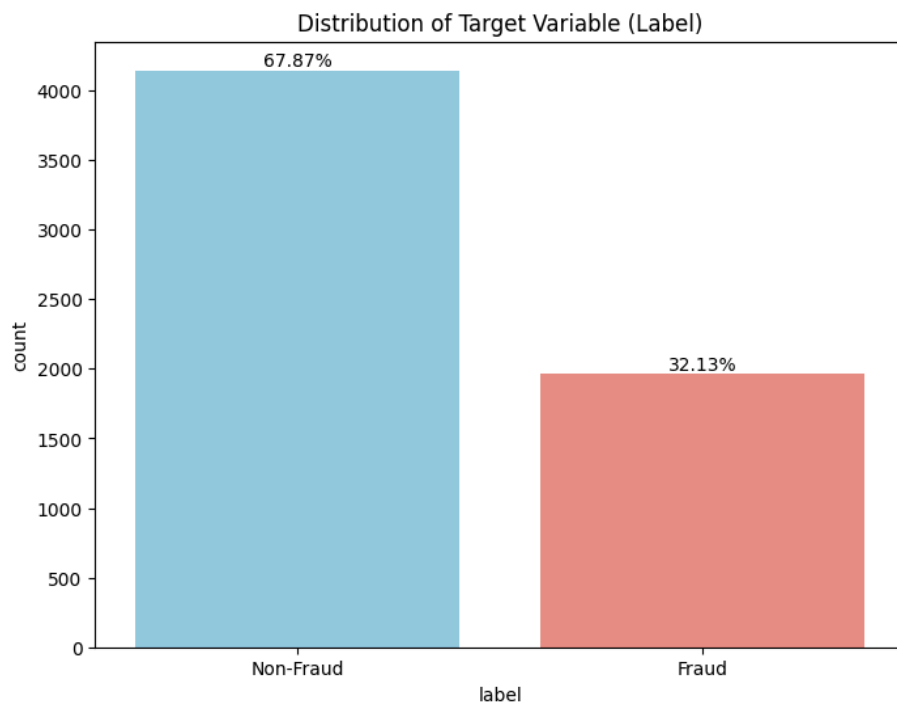


Fig 2: Label Distribution

3.3 Missing and Irregular Data Values

Both the App and User datasets had missing values, likely due to inactive phone numbers. In the App dataset, missing `month_id` and `busi_name` (app name) occurred for users without mobile data, while in the User dataset, missing ARPU (Average Revenue Per User) values across multiple months were likely due to plan suspension or deactivation. Imputation was performed after feature engineering to account for missingness patterns and ensure reliable statistics like variance and skewness (see Section 4.3).

We also observed negative ARPU values, typically indicating billing anomalies such as promotional rebates, refunds for service disruptions, or billing corrections.

4. Methods

4.1 Feature Engineering

To summarize user behavior more effectively, we aggregated app usage, SMS, and call data by `phone_no_m` across the four datasets. The table below outlines all engineered features, with detailed descriptions in the appendix (Section 9.1).

Data Source	Type of Feature Engineered	Feature Names
App Usage	App Usage	1. app_usage_count (Unique apps used)
	Data Flow Statistics	1. flow_mean 2. flow_median 3. flow_min 4. flow_max 5. flow_var 6. flow_sum 7. flow_month (average flow per month)
SMS Records	Volume and Diversity	1. sms_count (total SMS records), 2. sms_nunique_contact (unique contacts) 3. sms_date_nunique (unique days SMS was sent)
	Temporal Patterns	1. sms_hour_mode (most popular hour) 2. sms_day_mode (most popular day) 3. sms_hour_mode_count 4. sms_day_mode_count
		1. sms_hour_nunique (unique hours) 2. sms_day_nunique (unique days)
	Behavioural Ratios	1. sms_rate (total SMS count divided by unique contacts)
		1. sms_contacts_proportion (unique contacts divided by total SMS count)

	Call Type Proportions	1. sms_calltype1_count (outgoing sms count) 2. sms_calltype2_count (incoming sms count) 3. sms_calltype2_proportion (proportion of incoming SMS)
Voice Call Logs	Device, Contact and Geographic Diversity	1. imei_count (unique devices) 2. call_unique (unique contacts) 3. call_city_unique (unique cities) 4. call_county_unique (unique counties)
	Call Frequency	1. call_count (total calls) 2. call_count_mean 3. call_count_median 4. call_count_max (per contact)
	Call Duration Statistics	1. call_dur_mean 2. call_dur_median 3. call_dur_min 4. call_dur_max
		1. call_duration_per_contact_mean 2. call_duration_per_contact_median 3. call_duration_per_contact_max
	Temporal Patterns	1. voc_hour_mode (mode of call hour), 2. voc_day_mode (mode of call day) 3. voc_hour_mode_count 4. voc_day_mode_count 5. voc_dayname_mode_count
		1. voc_hour_unique (unique hours) 2. voc_date_unique (unique dates)
	Call Type Count and Proportions	1. voc_calltype1_count (outgoing calls count) 2. voc_calltype2_count (incoming calls count)

		3. voccalltype3_count (transfer calls count) 4. voc_calltype1_proportion (proportion of call type 1)
	Outgoing Call Geographic Diversity	1. call_outgoing_city_unique (unique cities for outgoing calls) 2. call_outgoing_county_unique (unique counties for outgoing calls)
User Consumption	ARPU (Average Revenue Per User) statistics	1. arpu_mean 2. arpu_var 3. arpu_max 4. arpu_min 5. arpu_median 6. arpu_sum 7. arpu_skew 8. arpu_sem (standard error of mean)
	Identity Verification	1. idcard_cnt (number of registered identity cards)

Fig 3: Feature Engineering Overview

4.2 Missing Data in the Aggregated Dataset: Identification and Imputation

4.2.1 Identification

The final aggregated dataset contains 6644 missing values, categorized into three groups based on data type and usage behavior:

Category	Count	Condition	Columns Affected
User and App Data	3929	Phone was not used throughout the eight months	All ARPU-related columns will be NA.
			flow_var will be NA (as all data flow values will be 0)
		Phone was only used for one month.	arpu_var, arpu_skew, and arpu_sem will be NA (due to insufficient data points for

			statistical calculation).
		Phone was only used for two months.	<code>arpu_skew</code> will be NA (due to insufficient data points for statistical calculation).
		Phone was used for more than two months	No missing values.
VOC Data	2664	Phone did not make or receive any calls throughout the 8 months	All VOC-related columns will be NA.
		Phone did not make any outgoing calls.	<code>call-outgoing_city_unique</code> and <code>call_outgoing_county_unique</code> will be NA.
SMS Data	51	Phone did not send or receive any SMS messages throughout the eight months.	All SMS-related columns will be NA.

Fig 4: NA Identification

4.2.2 Imputation

Imputing missing values with 0 is appropriate for most columns, as missing data usually indicates user inactivity rather than errors. For example, missing ARPU, call, or SMS records often reflect no activity. Filling with 0 preserves data integrity and accurately represents the lack of transactions. Therefore, we apply a simple fill-na with 0 across all columns.

4.3 EDA Insights

4.3.1 Insights

Exploratory data analysis revealed several interesting insights.

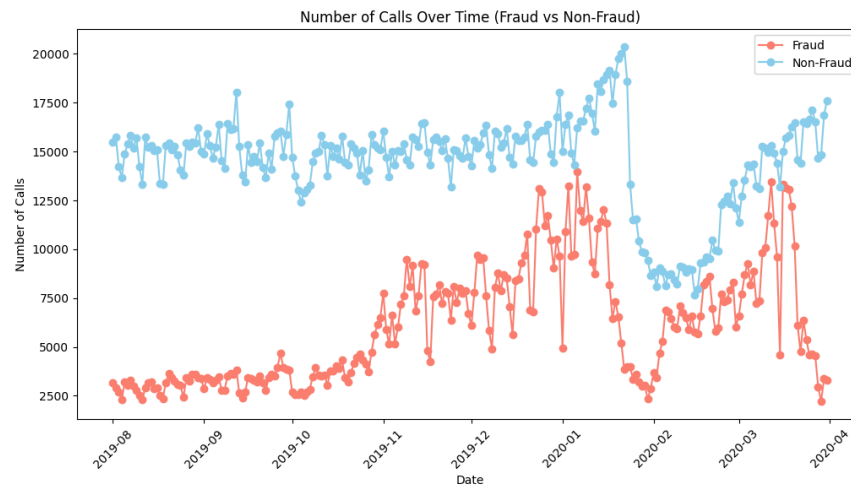


Fig 5: Fraudulent and Non-fraudulent Calls Distribution Over 8 Months

Fraudulent calls showed seasonality, rising from October 2019 and peaking in January 2020. This trend may be driven by the holiday season, where heightened gift-giving pressures leave consumers more vulnerable to scams offering deals or quick fixes (TransUnion, 2020). In contrast, non-fraudulent calls remained relatively steady, with an increase in January likely driven by Chinese New Year communication with loved ones.

Both call types dropped sharply after January, coinciding with COVID-19 disruptions and Chinese holiday travel (Jan 24–Feb 2). By April, non-fraudulent calls stabilized, possibly as people adjusted to the new normal. Scammers may have intensified efforts during the pandemic to exploit public anxiety and isolation, with the later decline in fraud calls potentially reflecting increased awareness, though this remains speculative.

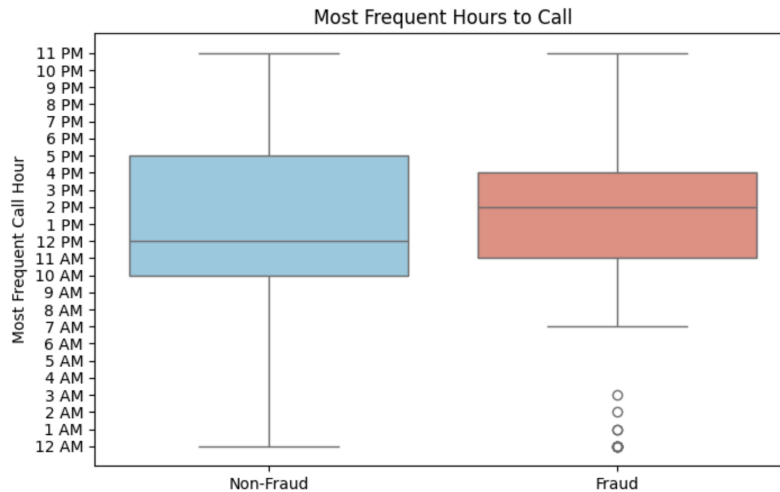


Fig 6: Boxplot of Call Timings for Fraudulent and Non-Fraudulent Numbers

Another significant finding is that fraudulent callers show peak activity between 10 AM and 4 PM, likely targeting times when recipients are most available. In contrast, non-fraudulent calls follow a broader 9 AM to 5 PM pattern. This suggests scammers may adopt a strategic approach, maximizing success by aligning with common call windows.

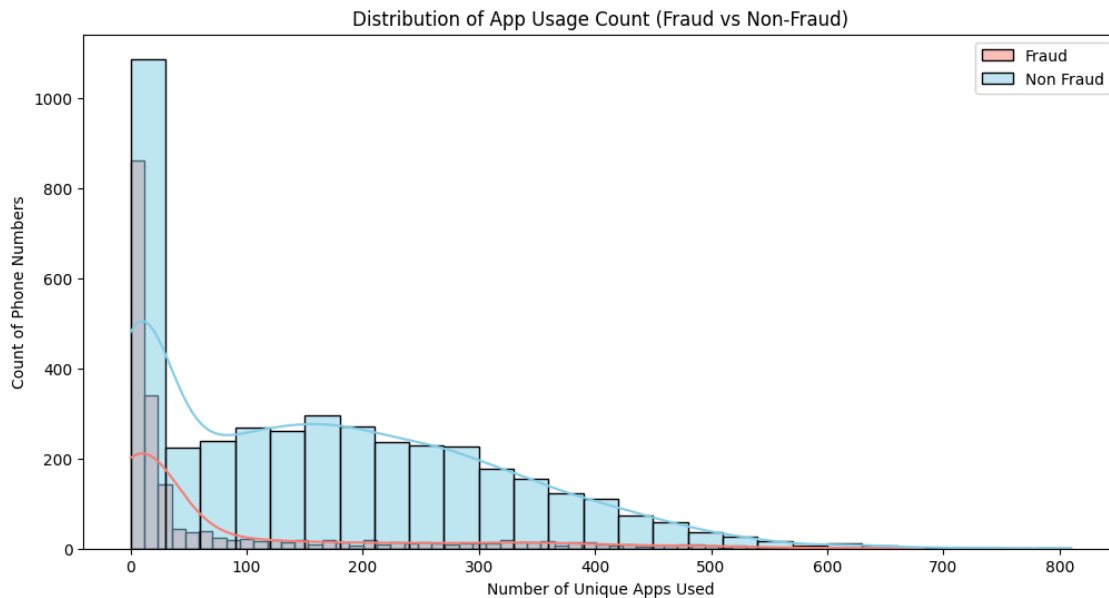


Fig 7: Distribution of the Count of Unique Apps Used for Fraudulent and Non-Fraudulent Numbers

It can also be noted that fraudulent numbers exhibit minimal to low app usage, primarily being used for making calls or messaging.

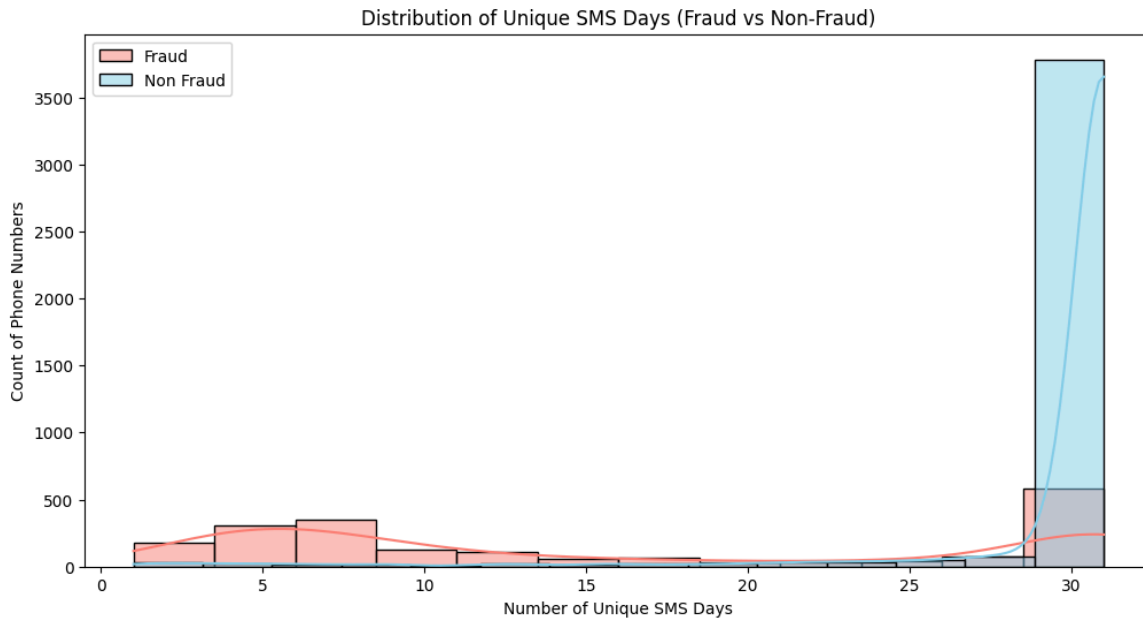


Fig 8: Distribution of Unique SMS Days for Fraudulent and Non-fraudulent Numbers

Lastly, fraudulent numbers typically send SMS on fewer unique days each month, concentrating their activity into short, high-intensity bursts rather than maintaining a steady daily pattern like non-fraudulent numbers. We believe this irregular behavior may be an attempt to evade detection while maximizing their reach.

4.3.2 Hypotheses

Hypothesis 1

We propose that fraudulent phone numbers are often associated with the same device, potentially due to fraudsters reusing devices, such as through phone cloning where multiple unauthorized phones are linked to a single IMEI, to evade detection (Gupta, 2024). The IMEI (International Mobile Equipment Identity) number is a unique 15-digit serial number for identifying a device.

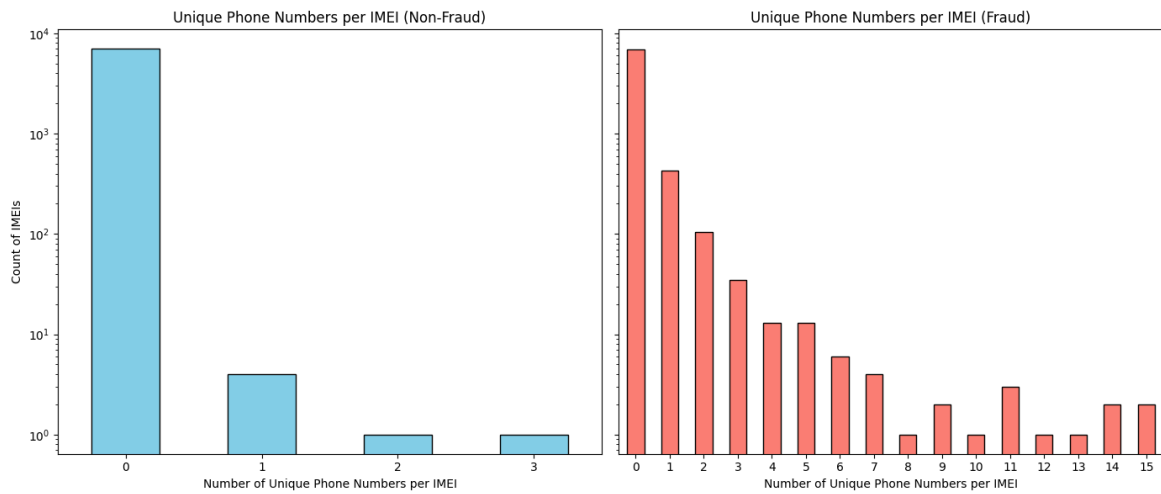


Fig 9: Count of Unique Phone Numbers Per IMEI Device

Conclusion: The above graph supports our hypothesis. Non-fraudulent numbers typically share an IMEI with three or fewer numbers, while fraudulent numbers can be linked to up to 15.

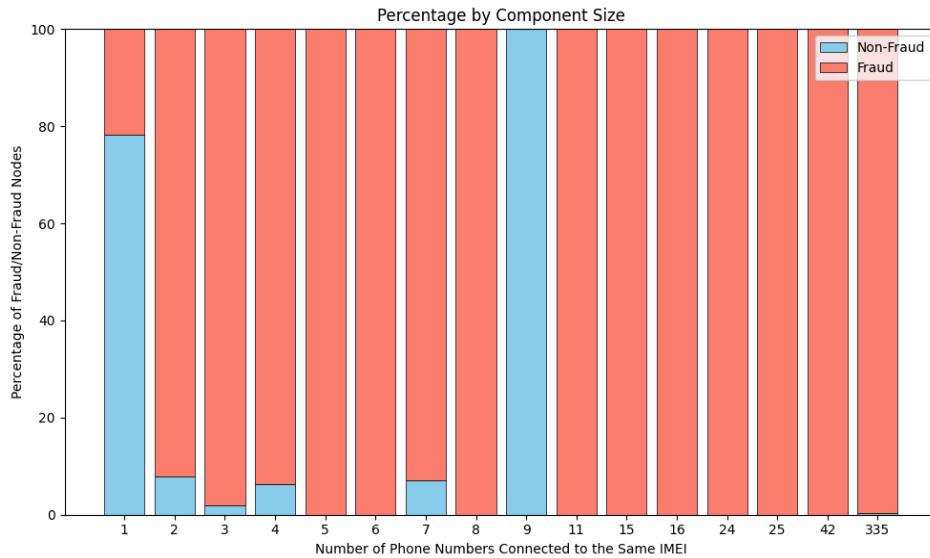


Fig 10: Percentage of Fraudulent and Non-Fraudulent Numbers Connected by the Same IMEI Device

We also constructed a graph (shown above) where each connected component represents phone numbers that share a device (IMEI). Fraudulent activity was more concentrated in larger components (2+ nodes), suggesting that multiple fraudulent numbers often operate from the same device. In contrast, single-node components had a lower fraud rate of 21.71%. Notably, the size-9 component with 0% fraud is an outlier reflecting legitimate multi-number use by a single user.

Implications: We plan to construct a graph where nodes represent phone numbers and edges reflect shared IMEI devices, which could improve fraud detection by identifying clusters of fraudulent users operating from the same device.

Hypothesis 2

We hypothesize that fraudsters are more likely to have multiple phone numbers registered under a single ID to facilitate impersonation or avoid number-level blocking (Ye & Bao, 2020). In contrast, legitimate users typically maintain only one or two numbers per ID for personal or professional purposes.

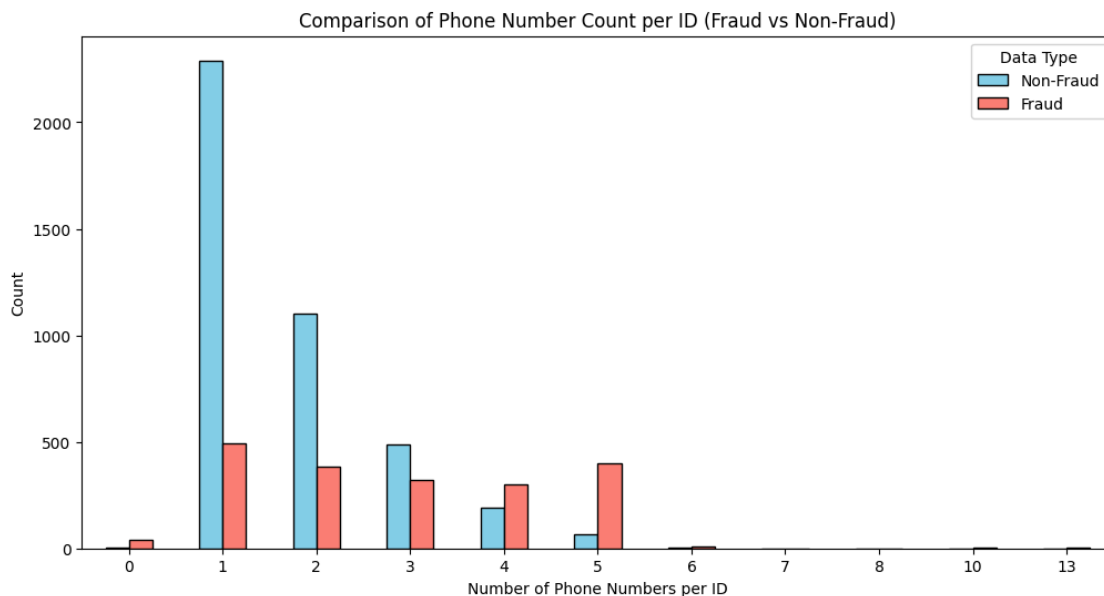


Fig 11: Number of Phone Numbers Per ID for Fraudulent and Non-Fraudulent Users

Conclusion: The data supports this. Non-fraudulent users usually have one or two numbers per ID, while fraudulent users often have two or more. Interestingly, some phone numbers—many of which are flagged as fraudulent—are not associated with any ID, possibly due to illicit acquisition or obtained prior to the implementation of identity registration regulations (Xinhua, 2015).

Implication: High or zero `idcard_cnt` can be useful indicators of fraud.

Hypothesis 3

We hypothesize that fraudulent phone numbers are often used across many devices, suggesting intentional reuse due to difficulties in legally obtaining new numbers under strict SIM registration laws (Xinhua, 2015). Legitimate users, by contrast, typically use their number on only a few personal devices. Additionally, prior investigations into device tracking in China showed that some manufacturers reuse IMEI numbers across devices (OpIndia Staff, 2020), which may introduce noise.

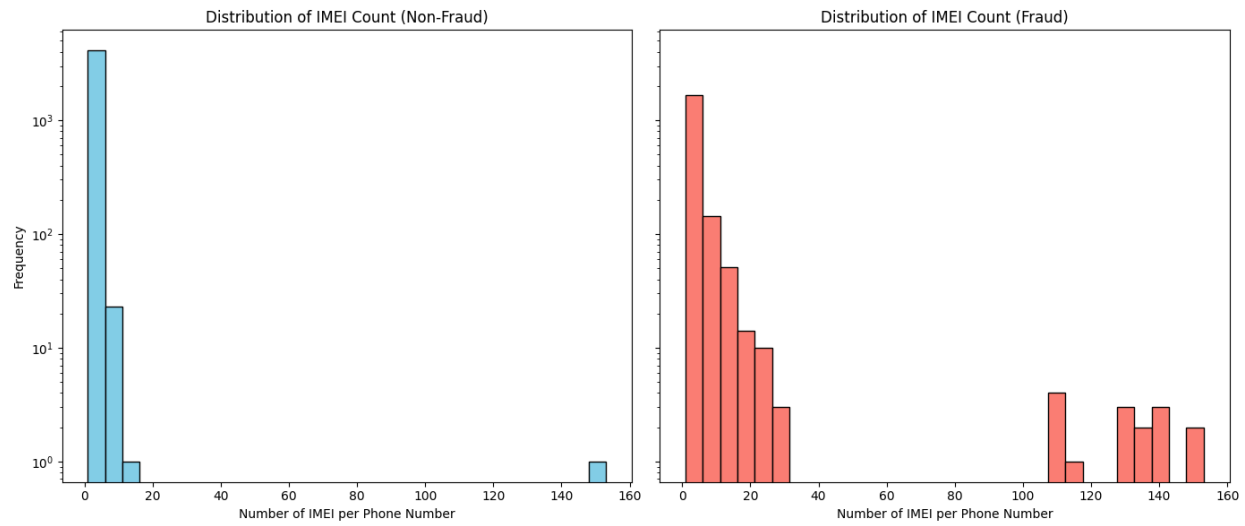


Fig 12: Distribution of the Number of IMEI Devices used Per Phone Number

Conclusion: Non-fraudulent numbers usually appear on a few IMEIs, while some fraudulent ones can be linked to over 100.

Implication: We will engineer a new feature, `imei_count`, which captures the number of unique IMEI devices per phone number for fraud detection.

Hypothesis 4

We hypothesize that Fraudulent numbers likely have a skewed outgoing-to-incoming call ratio, as scams are typically one-sided. In contrast, non-fraudulent users should display more balanced, two-way communication (Jiang et al., 2013).

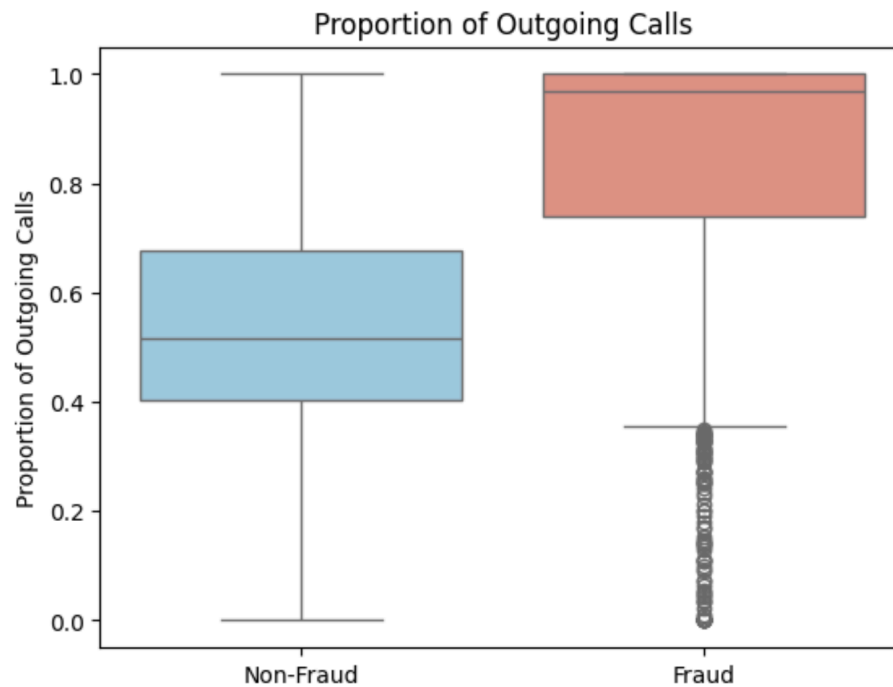


Fig 13: Boxplots of Proportion of Outgoing Calls for Fraudulent and Non-Fraudulent Phone Numbers

Conclusion: Our analysis supports the hypothesis. The chart above demonstrates that fraudulent numbers have outgoing call ratios near 1, indicating they predominantly make outgoing calls. Conversely, non-fraudulent numbers maintain a more balanced ratio around 0.5, reflecting typical two-way communication.

Implications: We create a new feature, `calltype1_proportion`, to capture the ratio of outgoing to total calls for detecting fraud.

Hypothesis 5

Fraudulent numbers are expected to contact more unique phone numbers than non-fraudulent ones, as they aim for wider reach with fewer repeated interactions. In contrast, non-fraudulent users tend to communicate within a smaller, familiar network of friends, family or colleagues (Jiang et al., 2013).

	label	count	mean	std	min	25%	50%	75%	max
0	Non-Fraud	4133.0	114.554319	187.688450	1.0	27.0	61.0	137.00	3731.0
1	Fraud	1892.0	455.422304	1252.626616	1.0	56.0	138.0	307.25	13746.0

Fig 14: Statistics of Count of Unique Phone Numbers Called

	label	count	mean	std	min	25%	50%	75%	max
0	Non-Fraud	4141.0	166.313209	208.173929	1.0	36.0	92.0	226.0	5808.0
1	Fraud	1962.0	344.338430	1453.253247	1.0	8.0	18.0	128.0	20931.0

Fig 15: Statistics of Count of Unique Phone Numbers Messaged

Conclusion: The data supports our hypothesis. Fraudulent numbers contact a significantly higher mean number of unique contacts, in their preferred channel of either calls, messages, or both, compared to non-fraudulent numbers.

Implications: We will engineer four new features: `call_unique` (unique numbers contacted over eight months), `call_count_max` (maximum number of calls exchanged with one number), `sms_nunique_contact` (unique SMS contacts), and `sms_rate` (average number of messages per unique contact). Additionally, we will construct a graph where nodes represent phone numbers and edges represent communication links via calls or SMS for usage in our models. This could help identify clusters of non-fraudulent users and detect isolated fraudulent behaviours.

Hypothesis 6

We hypothesize that fraudulent numbers exhibit greater variation in call durations, with scams leading to abrupt or unusually long calls. In contrast, non-fraudulent phone numbers tend to have more consistent and moderate call durations (Jiang et al., 2013).

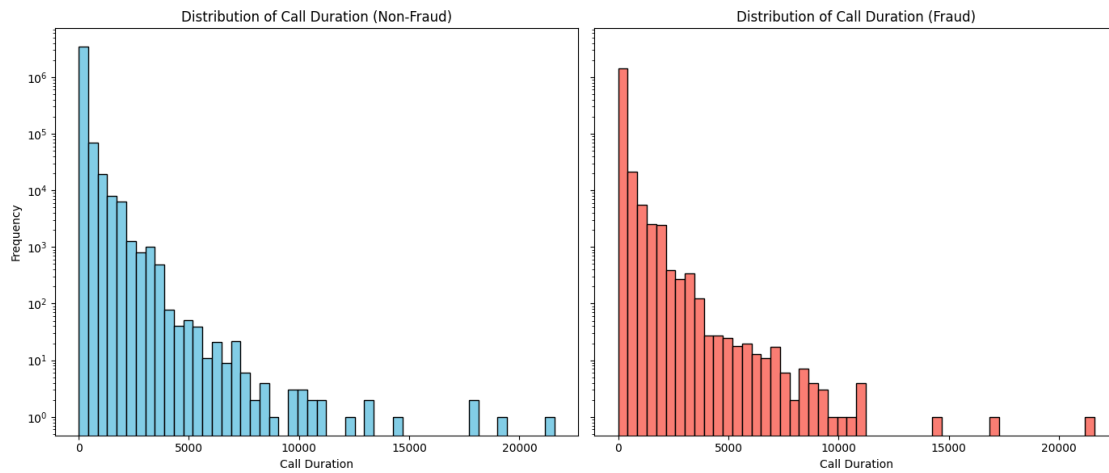


Fig 16: Distribution of Call Duration for Fraudulent and Non-Fraudulent Numbers

Conclusion: Our analysis does not prove this hypothesis as the call duration distribution for both fraudulent and non-fraudulent numbers are quite similar. Interestingly, fraudulent numbers have a minimum call time of 0, but not for non-fraudulent numbers.

Hypothesis 7

We hypothesise that fraudulent users often display sporadic phone usage, leaving their SIM cards inactive for extended periods. This behaviour could be reflected in lower or zero data flow and ARPU values, where SIMs are only activated when needed.

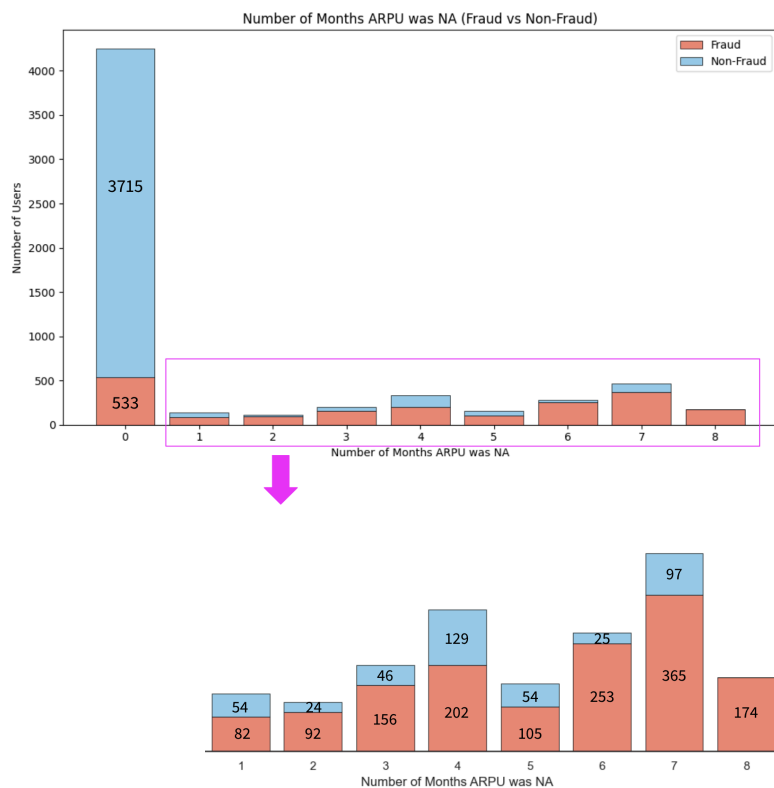


Fig 17: Distribution of Users by Number of NA Months in ARPU

Conclusion 7: Our findings support this hypothesis. While most fraudulent and non-fraudulent users have 0 months of missing ARPU, suggesting consistent usage, a notable divergence appears among users with 1 to 8 months of missing ARPU. From month 4 onwards, the proportion of fraudulent users increases steadily compared to non-fraudulent users. This trend suggests that fraudulent users are more prone to prolonged inactivity. Conversely, non-fraudulent users tend to have no months with missing ARPU values, suggesting more consistent usage.

Implication 7: Based on this finding, we will input the missing ARPU values with 0 and engineer new features, `arpu_var`, `arpu_skew` and `flow_var`, which capture the variance in ARPU and data flow. These features will help identify fraudulent users by highlighting irregular usage patterns and periods of inactivity.

4.4 Feature Selection

To address multicollinearity in the final aggregated dataset, we calculated feature correlations and dropped those with a value above 0.95, ensuring that only highly correlated features are removed, avoiding unnecessary column drops and potential information loss.

The dropped features are listed below.

Feature Dropped	Correlated With	Correlation Score
voc_dayname_mode_count	call_count	0.984299
sms_dayname_mode_count	sms_day_mode_count	0.980084
flow_sum	flow_month	0.972392
call_county_unique	call_outgoing_county_unique	0.967204
voc_hour_mode_count	voc_dayname_mode_count	0.959552
sms_day_mode_count	sms_nunique_contact	0.953830

Fig 18: Feature dropped based on correlation scores above 0.95

To tackle noisy features, we performed logistic regression solely for feature selection before graph construction. The model was trained on 80% of the data, and feature importance was assessed based on the absolute values of the coefficients, $|\beta_i|$. Predictors with $|\beta_i|$ below a threshold τ (tested between 0.05 to 0.35) were discarded. Importantly, the coefficients were not used in the final model, eliminating any risk of data leakage. Therefore, this step is designed solely to filter out less informative features and enhance downstream performance.

Threshold	No of Predictors Kept (initial:54)	Overall Accuracy	Recall (Fraud)
0.05	46	0.914	0.810
0.10	40	0.922	0.823
0.15	37	0.924	0.823
0.20	32	0.927	0.835
0.25/0.30	28	0.935	0.848
0.35	26	0.935	0.835

Fig 19: Performance metrics obtained from our experimental approach based on ensemble model combining GNN and Random Forest (Refer to Section 6.2.2)

Ultimately, we selected a feature selection threshold of 0.25, balancing model simplicity and performance, with an emphasis on recall for fraudulent cases. This threshold reduced predictors from 54 to 28, achieving the highest recall (0.848) and strong accuracy (0.935). This is especially important in fraud detection tasks, where recall—the ability to correctly identify fraudulent cases—takes precedence over precision, without compromising too much on overall accuracy.

Although a threshold of 0.35 yields similar performance (recall 0.835, accuracy 0.935), it retains only 26 predictors. As such, we chose 0.25 to ensure a richer feature set for graph-based analyses, balancing predictive strength with interpretability.

5. Graph Construction

5.1 Similarity Graph

We construct an undirected weighted graph where nodes represent phone numbers and edge weights are determined by the cosine similarity of their features. Nodes with higher similarity are connected, aiding in the detection of fraud rings versus non-fraudulent networks. Using a k-nearest neighbor (k-NN) approach, we ensure each node has at least k connections, reducing isolation and maintaining sufficient graph connectivity for optimal model performance. As noted by Zhang et al. (2019), maintaining sufficient connectivity in similarity-based graphs is crucial for effective learning. Their work demonstrates that well-connected graphs enable better message passing and richer node representations, which we aim to achieve in fraud detection.

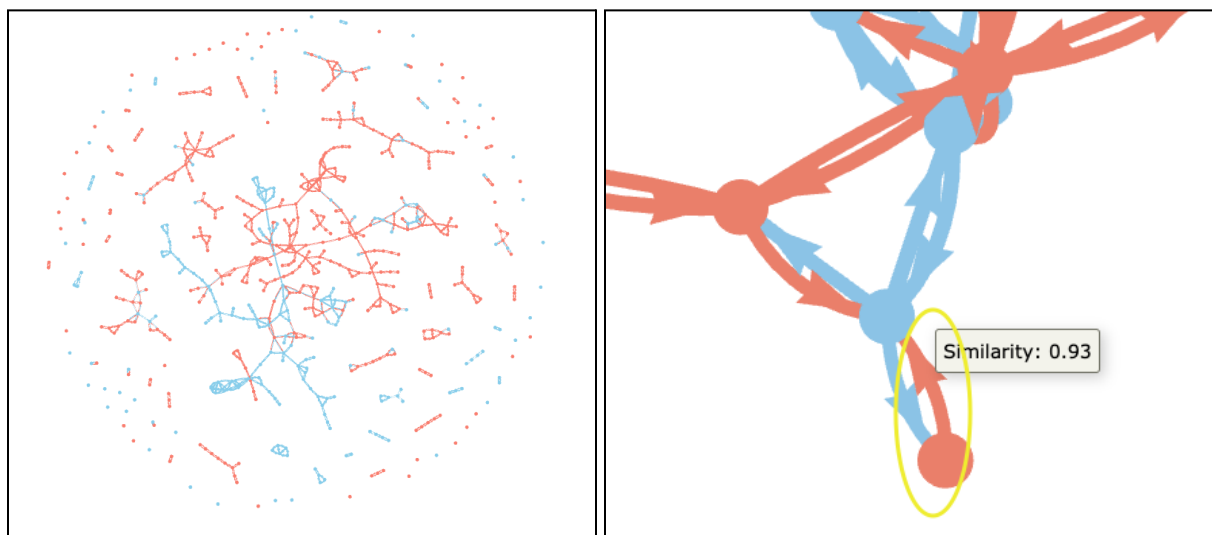


Fig 20 (left): A visualisation of a **subset** of our graph containing 1,000 nodes. Some nodes appear isolated because all their 15 nearest neighbors fall outside this subset—in the remaining 5,000 nodes not shown here.

Fig 21 (right): A zoomed-in view of a graph segment, where **red nodes** indicate fraud cases and **blue nodes** represent non-fraud cases. **Edges** between nodes display their corresponding **similarity scores**.

Moreover, our use of a k-NN approach aligns with Anastasiu and Karypis (2015), who found that moderate k-values (10–50) strike a balance between preserving graph structure and controlling computational costs. Limiting neighbors to the most similar nodes fosters local cohesion and distinct clusters which is particularly relevant in our context of identifying similar behavioral patterns among phone numbers.

However, too many neighbors can introduce noise, as noted by Quan et al. (2023), who found that excess connections can obscure useful patterns and reduce model accuracy. To mitigate noise and ensure relevance, we limit testing each node's neighbors to 50. This choice also aligns with Ruiz et al. (2023), who showed that GNNs perform best with balanced graph sparsity and connectivity. Additionally, limiting neighbors addresses the potential issue of scalability in large networks. With no more than 300,000 edges for 6,000 nodes, we maintain efficient training without sacrificing representational power.

Motivated by these insights, we then experimented with small k-values of 10, 15, 20, and 50 to identify the optimal value for our model.

K-Value	Overall Accuracy	Recall (Fraud)	F1-score (Fraud)
10	0.931	0.848	0.887
15	0.935	0.848	0.893
20	0.927	0.848	0.882
50	0.918	0.835	0.868

Fig 22: Performance metrics obtained from our experimental approach on ensemble model combining GNN and Random Forest (Refer to Section 6.2.2)

We selected $k = 15$ as it offers the best trade-off across metrics, achieving the highest accuracy of 0.935 and a strong F1-score of 0.893, while maintaining stable recall around 0.848. Increasing k beyond 20 led to diminishing returns, with performance dropping at $k = 50$, making **$k = 15$** the optimal choice for both effectiveness and generalization.

5.2 Shared IMEI Graph

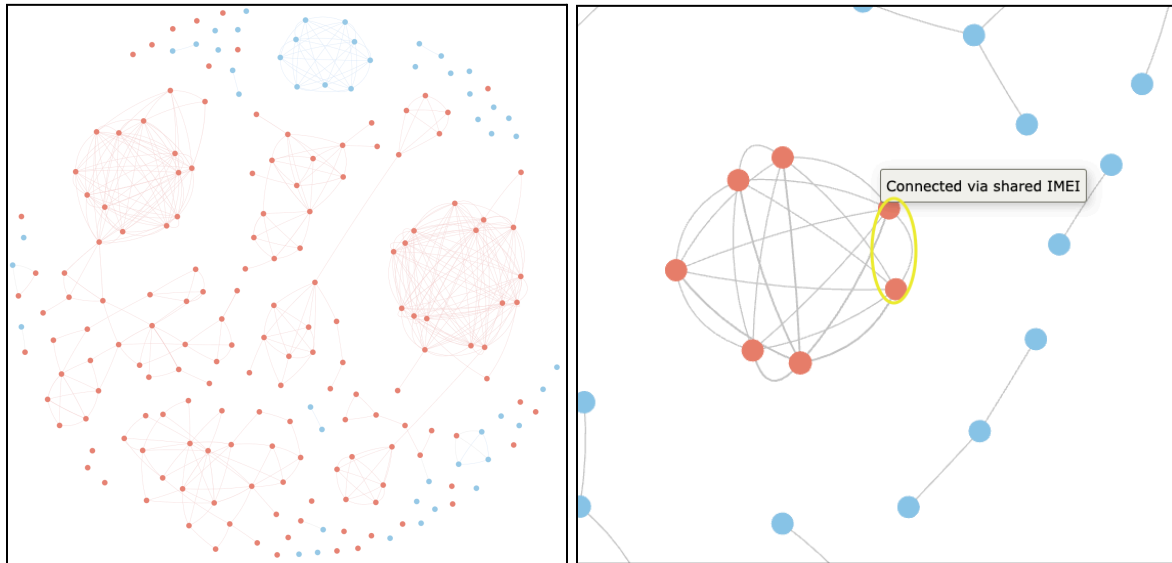


Fig 23 (left): A visualisation of a subset of our graph containing 600 nodes. Only nodes with relations in the subset were included for the visualisation.

Fig 24 (right): A zoomed-in view of a graph segment, where red nodes indicate fraud cases and blue nodes represent non-fraud cases. The edges between nodes also display their relationship based on shared IMEI.

For the Shared IMEI Graph, nodes represent phone numbers with edges connecting numbers that used the same IMEI device. It was formed based on hypothesis 1, which suggests fraud often occurs in groups. Most non-fraudulent phone numbers appear as single nodes or are connected to only a few others, while many fraudulent phone numbers appear in larger connected components.

5.3 VOC and SMS Interactions Graph

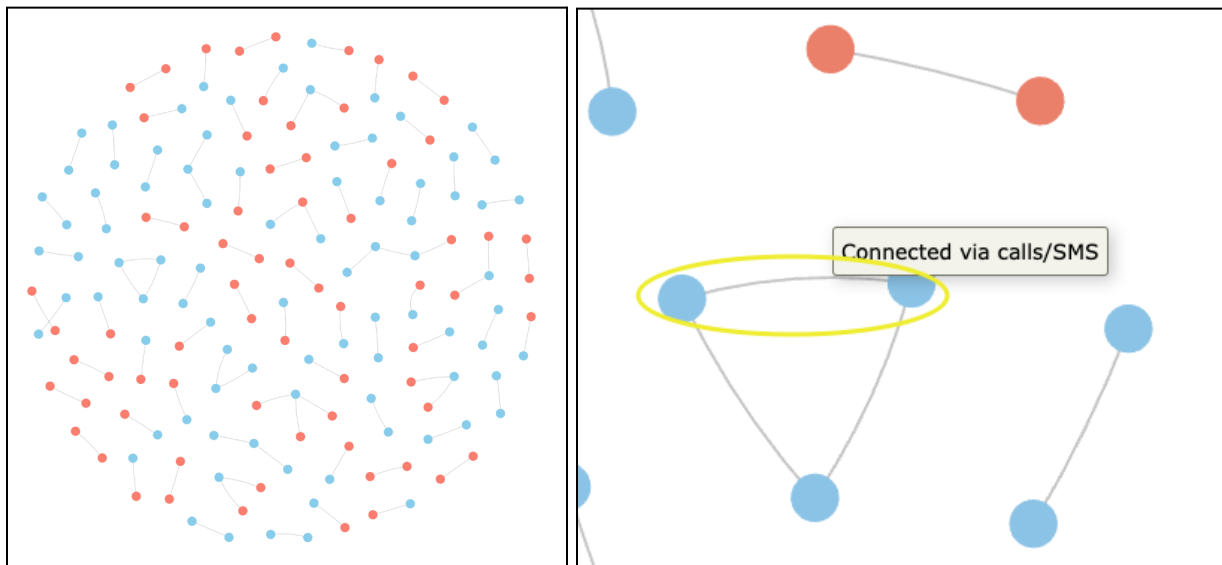


Fig 25 (left): A visualisation of a subset of our graph containing 600 nodes. Only nodes with relations in the subset were included for the visualisation.

Fig 26 (right): A zoomed-in view of a graph segment, where red nodes indicate fraud cases and blue nodes represent non-fraud cases. The edges between nodes also display their relationship based on call/sms.

This graph visualizes user interactions via calls or SMS. Each node represents a phone number, and edges are drawn when two numbers within our dataset exchange calls or messages. The graph structure reflects Hypothesis 5, which proposes that non-fraudulent users usually interact within smaller, familiar circles—likely made up of other non-fraudulent individuals. This pattern is evident in our plotted data, which seems to support the hypothesis. Additionally, we observed that fraudulent users often connect with other fraudsters, possibly to coordinate activities or exchange information (DataVisor, 2024).

However, although we have records of the numbers our users contact, only a limited subset of those interactions involve other users in our dataset. As a result, most connections lead to external numbers for which we lack comprehensive data, contributing to a relatively sparse interaction graph.

6. Models

6.1 Baseline Model - Relational Neighbour Classifier (RNC)

The RNC estimates class probabilities solely based on entities of the same type, whose class labels are known (Provost, 2019). This makes it easy to implement and provides insight into how local neighbourhood labels influence classification. On the assumption of homophily in fraud detection where fraudulent numbers exhibit similar behaviours and non-fraudulent numbers are similar to other non-fraudulent numbers, RNC was applied to the similarity graph. In determining the classification of a node, RNC takes the class with higher probability, between the probability of node being fraud and probability of node being non-fraud, based on the labels of neighbours in a predefined neighbourhood.

6.2 Challenger Models

Models attempted in the preliminary stage (listed in appendix) could only be applied to one graph each time. For consistency in comparing models, the similarity graph was the default to train models. However, as mentioned in Section 5 Graph Construction, we have identified 3 relationships (supported by findings in Section 4 Exploratory Data Analysis) useful in fraud classification. To make full use of the dataset, CARE-GNN was implemented as it could aggregate multiple graph relationships for classification. Besides CARE-GNN, the Feature Ensemble Method was implemented in an attempt to increase predictive performance by aggregating predictions generated by multiple Graph Neural Networks (GNN).

6.2.1 CARE-GNN model

CARE-GNN introduces mechanisms to resist feature camouflage where fraudulent nodes mimic features of non-fraudulent ones to evade detection, and relation camouflage where fraudsters establish connections with non-fraudulent ones, making their behaviour appear normal within the network (Dou et al., 2020). We adapted the CARE-GNN architecture to fit our dataset, designing it to enhance the model's ability to focus on relevant information and neighbours and to incorporate information across different types of relationships, allowing diverse interaction patterns to be captured.

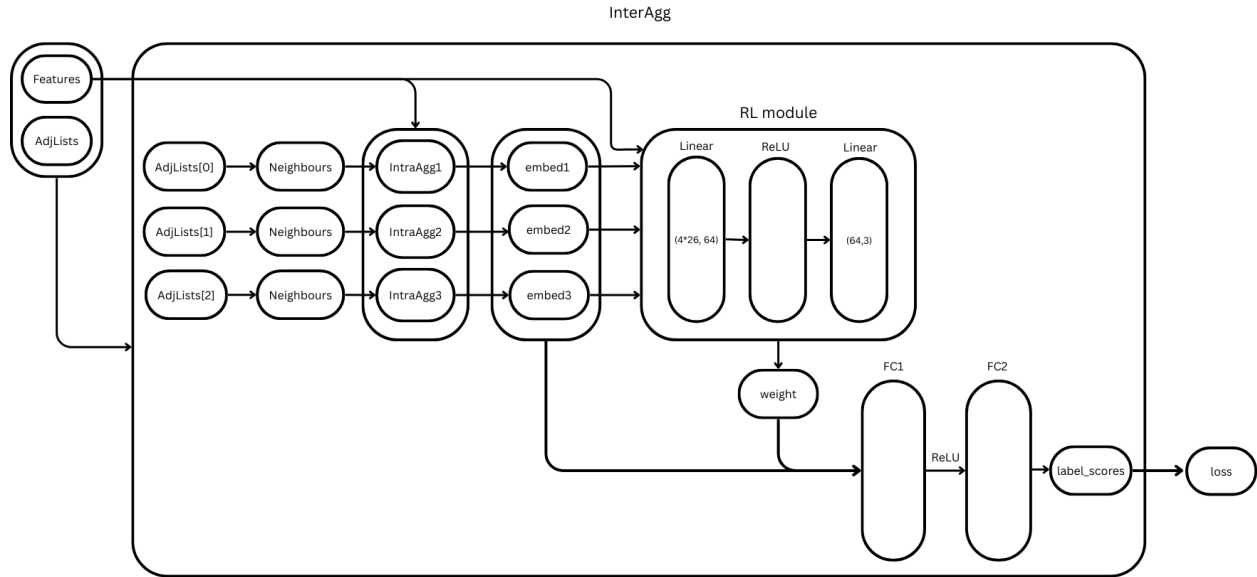


Fig 27: CARE-GNN Model Architecture

For each node, the model aggregates information from a sample of neighboring nodes for each relation, creating separate relation specific feature embeddings. Embeddings and original features were firstly used to compute relationship weights for the model to learn relative importance of relations for the classification of the node. Then embeddings and features were concatenated with the learned relation layer weights to form a final node embedding. The final node embedding is used in classification layers to generate scores and class.

6.2.2 Feature-Level Ensemble Model¹

Inspired by successful applications in fields like optical remote sensing—where fusing visual descriptors from multiple sources enhances detection accuracy (Feature Fusion – an Overview | ScienceDirect Topics, n.d.)—we propose a feature-level ensemble method that merges the strengths of multiple Graph Neural Network (GNN) architectures: GCN, GAT, and GraphSAGE. Instead of relying on a single model, we combine their embeddings and feed the unified representation into a Random Forest (RF) classifier.

Each GNN contributes a unique perspective: GCN captures global structural information, GAT assigns attention-based weights to neighboring nodes, and GraphSAGE samples neighborhoods to boost generalization and prevent overfitting. Fusing these embeddings produces a richer representation, capturing both global and local patterns while mitigating class imbalance by amplifying subtle signals from minority classes like fraudulent users, that could be drowned out in single-model settings.

¹ Used for our experimentation to determine the k-value (Section 5.1) and threshold value (Section 4.4), ensuring that we do not select values that could cause overfitting in our best model, the Care-GNN model.

This hybrid method, combining GNNs as feature extractors with an RF classifier for final prediction, integrates the structure-aware learning of GNNs with the interpretability and stability of classical machine learning, offering a scalable, effective solution for fraud detection.

7. Models

7.1 Training-Testing Approach and Chosen Metrics

For our RNC model, we used a 75–25 train-test split via scikit-learn's default function. In determining the classification of nodes from the test set, only labels of neighbour nodes in the training set were used to calculate class probabilities.

The CARE-GNN model employed a stratified split into training (64%), validation (16%), and test (20%) sets to preserve class distribution. The model was trained exclusively on the training set, while classification accuracy, AUC and cross-entropy loss were computed on the validation set at the end of every epoch to guide early stopping. Early stopping was implemented to prevent overfitting, halting training if the validation loss failed to improve for a specified number of consecutive epochs, thereby enhancing generalization and ensuring model performance.

Whereas, for training our feature ensemble model, each GNN was applied to the similarity graph, with the loss calculated based on 4,800 training nodes from an 80-20 train-validation split. Embeddings were generated for all ~6,000 nodes, preserving each model's unique insights. To evaluate performance, an additional 80-20 split was applied to the remaining ~1,200 nodes, reserving 250 for final testing with the RF classifier. To reduce variance from the small test set of ~250 nodes, we ran the RF classifier across 100 random seeds with different train-test splits. This approach improved metric reliability and reduced sensitivity to specific samples. By holding out data at both stages and averaging results, we ensured strong generalization and minimized overfitting.

We have selected Area Under the ROC Curve (AUC) as our primary evaluation metric, as it captures the model's ability to distinguish between fraudulent and legitimate calls across all classification thresholds (Ling et al., 2023). In telecom fraud detection, recall is prioritized over precision as missing a fraudulent call (false negative) poses greater financial and reputational losses (Shen et al., 2025), as compared to false positives. AUC allows flexible threshold adjustments to optimize for high recall while maintaining reasonable precision.

7.2 Model Results

Algorithm	Recall (Fraud)	AUC
RNC (Baseline Model)	0.75	0.8660
Feature-Level Ensemble (GCN,GAT, Graphsage node embeddings + Random Forest to train embeddings)	0.84	0.9420
CARE-GNN	0.84	0.9477
CARE-GNN (fine-tuned)	0.86	0.9534

Fig 28: Overall comparison of model performance for baseline model and challenger models

Our baseline model, the RNC, achieved a recall of 0.70 and an AUC of 0.8660, establishing a solid benchmark for assessing the performance of more advanced approaches.

Confusion Matrix (RNC)		
	Pred 0	Pred 1
Actual 0	1004	21
Actual 1	124	377

Fig 29: Confusion Matrix for RNC Predictions

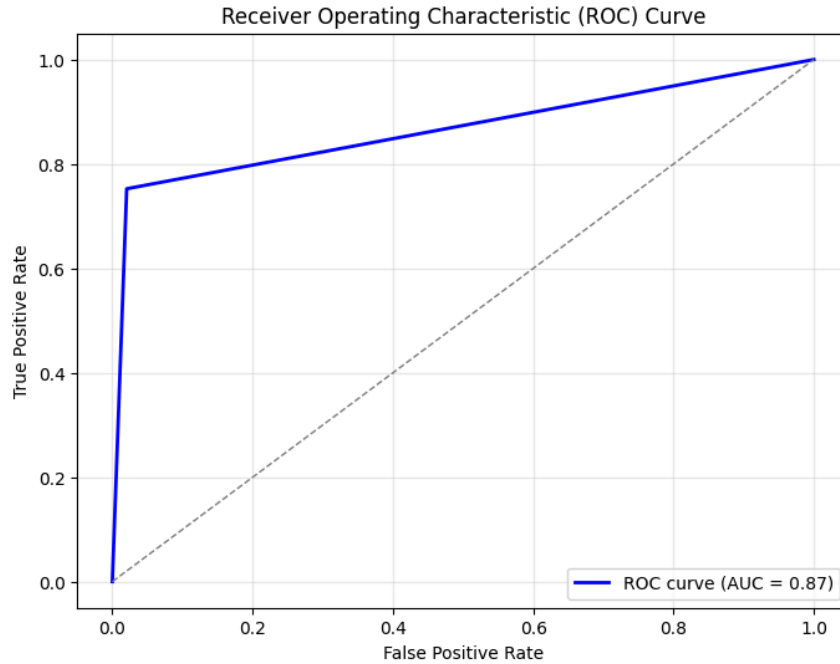


Fig 30: Receiver Operating Characteristic (ROC) Curve on RNC Predictions

The Feature-Level Ensemble Model, which integrates node embeddings from GCN, GAT, and GraphSAGE into a RF classifier, substantially outperformed the baseline. It attained a recall of 0.84 and an AUC of 0.9420, demonstrating the effectiveness of leveraging multiple graph-based representations to capture complex node relationships relevant to fraud detection.

Average Confusion Matrix Over 100 Runs (Feature Ensemble)		
	Pred 0	Pred 1
Actual 0	162	4
Actual 1	12	67

Fig 31: Averaged Confusion Matrix for Feature-Level Ensemble Model Predictions

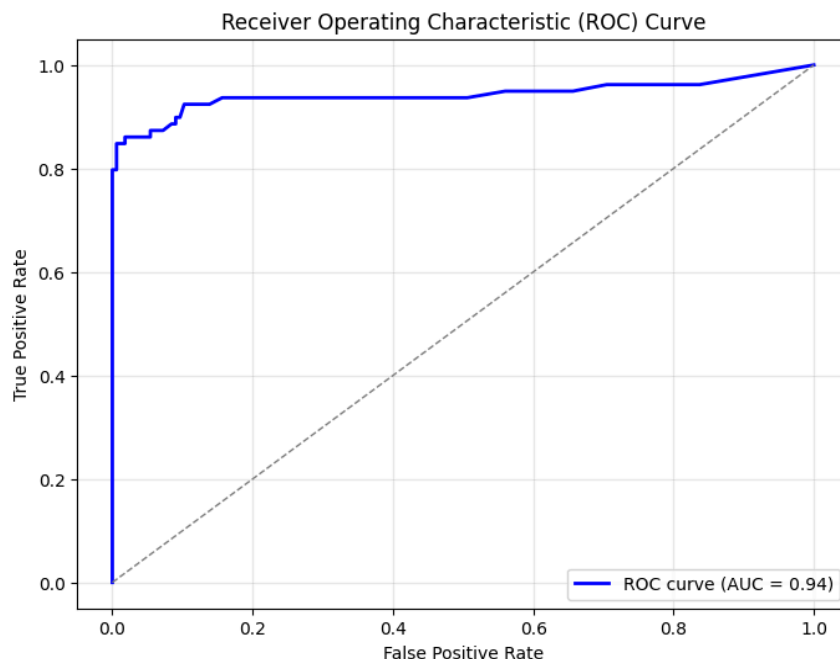


Fig 32: Receiver Operating Characteristic (ROC) curve on Feature-level Ensemble Model predictions

Building on this, CARE-GNN delivered the best performance, achieving a recall of 0.84 and an AUC of 0.9477, having better AUC over its challenger Feature-Level Ensemble model. Its context-aware relational learning mechanism enables the model to effectively differentiate subtle patterns indicative of fraudulent behavior across heterogeneous edge types.

Given our objective of maximizing recall to reduce the risk of undetected fraud, while also maintaining strong discriminative power as reflected by the AUC, CARE-GNN was selected as the final model.

7.3 Hyperparameter Tuning via Grid Search

To optimize our final CARE-GNN model performance, we conducted hyperparameter tuning using a grid search approach. Model evaluation was based on the validation AUC to ensure robust generalization.

Key hyperparameters considered during tuning included:

- **Learning Rate:** Controls the step size during optimization. A high learning rate can lead to unstable training, whereas a low rate may slow convergence.
- **Regularization Strength:** Regulates the balance between overfitting and underfitting by influencing the contribution of the relation-weighted loss term.
- **Dropout Rate:** Mitigates overfitting by randomly deactivating neurons during training, requiring careful calibration to avoid compromising model capacity.

The grid search revealed that optimal tuning of these parameters led to a 2% improvement in recall for fraud detection using CARE-GNN.

Confusion Matrix (fine tuned CARE-GNN)		
	Pred 0	Pred 1
Actual 0	785	44
Actual 1	60	333

Fig 33: Confusion Matrix for Fine Tuned CARE-GNN Predictions

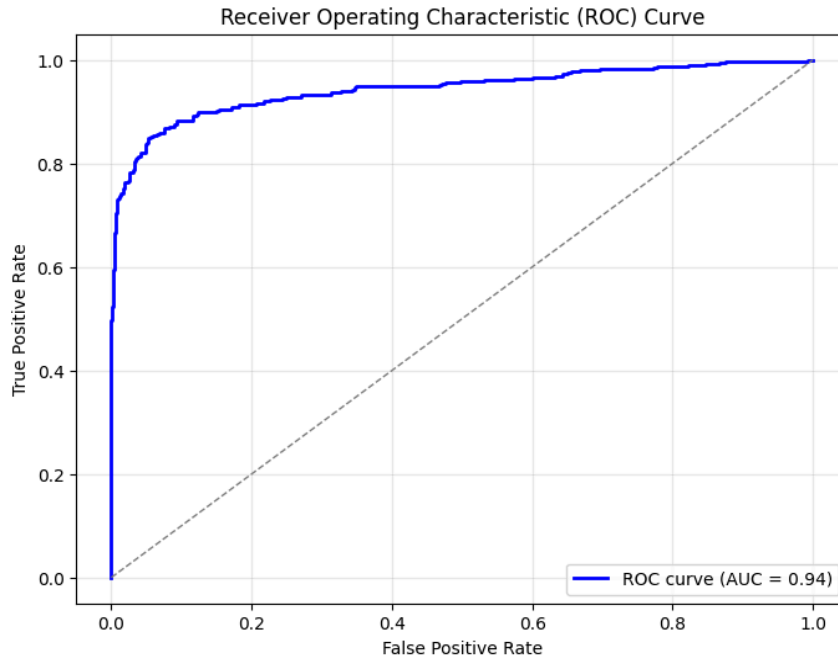


Fig 34: Receiver Operating Characteristic (ROC) Curve on Fine Tuned CARE-GNN Predictions

7.4 Explainability Tools On CARE-GNN

CARE-GNN predictive behaviour would be explored in two aspects, structural (relation) importance and feature importance.

7.4.1 Relationship Importance

As CARE-GNN incorporates 3 different relationships into model predictions, model predictions can be more interpretable and transparent by identifying the relative importance of the various relationships in determining the class for a particular phone number. This can be easily retrieved by indexing the `relation_weights_global` matrix that keeps track of updated relationship weights. We were interested to identify if there were particular relationships that were of higher importance across all nodes.

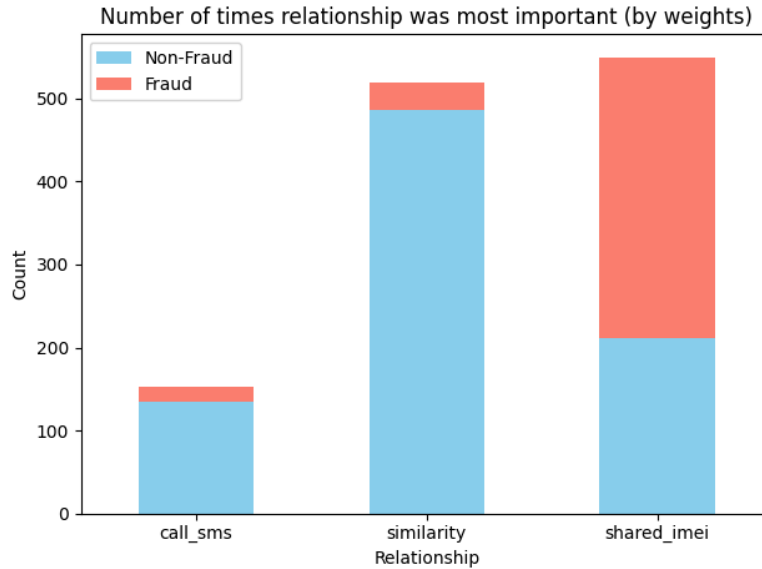


Fig 35: Distribution of Count of Relationship being Most Important by Weights

Based on the bar graph, the similarity relationship between phone numbers is the most important relationship for identifying non-fraud numbers. This is likely because non-fraud numbers tend to behave similarly, whereas fraud numbers exhibit more varied behaviors due to differing scam styles (observed in Section 4.3).

For predicting phone numbers as fraud, shared IMEI is the most important relationship. This supports Hypothesis 1, where we found fraud numbers are more likely to share the same device. Incorporating this into CARE-GNN improved model performance and recall, helping to identify fraud based on operational characteristics.

In contrast, call and SMS interactions are least useful for predicting both classes, likely because they lack class-specific patterns, and the graph's sparsity limits its effectiveness. When such links exist, they're more effective for non-fraud detection, likely because non-fraud users call known contacts—unlike fraud users, who rarely connect with each other.



Fig 36: Relation Weight Trends during Training

To supplement the relative importance of various relationships in predicting phone numbers as fraudulent or non-fraudulent, this graph shows changes in relation weights (higher weights, higher importance of the relation) as CARE-GNN was trained. In the initial steps, average weights of the relationships fluctuate, possibly due to the model still learning various patterns in the input data and adjusting initial weights towards more accurate predictions. From the middle stage of training, average weights for all relationships were relatively stable, with `shared_imei` having highest average weight (most important across all numbers), followed by similarity and calls.

7.4.2 Feature Importance

To gain a deeper understanding of how CARE-GNN predicted fraud or non-fraud for each phone number (i.e., instance-level explanations), we employed model-agnostic explainability tools, including SHapley Additive exPlanations (SHAP) (Lundberg, 2023) and Integrated Gradients (IG) with Captum (Sundararajan et al., 2017).

7.4.2.1 SHAP

SHAP was used to quantify feature importance for model interpretability. However, SHAP is not directly applicable to complex GNNs like CARE-GNN due to their non-Euclidean structure. Thus, we adopted a model-agnostic surrogate approach (Saha, 2022) by training a simple and interpretable surrogate model, RF, to approximate the predictions of the CARE-GNN. The

surrogate model was trained with the same inputs as the original CARE-GNN, with the target being the CARE-GNN's predicted outputs. This enables the use of SHAP to generate reliable feature importance scores. While we acknowledge that this approach may not fully capture the relational inductive biases learned by the GNN, it provides a practical and interpretable approximation of individual feature contributions for CARE-GNN.

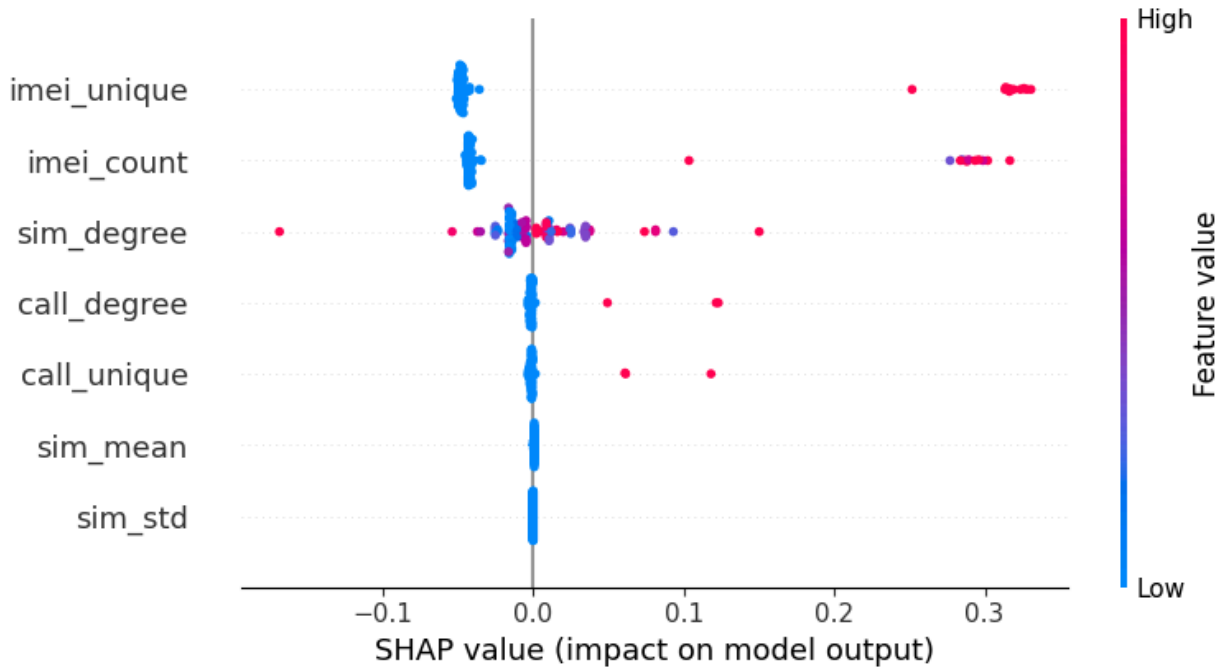


Fig 37: SHAP Beeswarm Plot using Surrogate Model

The above plot reveals the relationships of individual instances (phone numbers) to being predicted as fraud. For features mentioned in the plot, higher feature values tend to contribute towards predicting numbers as fraud and lower feature values either contribute negatively or have minimal impact. By mean absolute SHAP values across all phone numbers, `imei_unique` has the highest mean absolute SHAP value and is ranked most impactful on model predictions, followed by `imei_count`, and `sim_std` (standard deviation for similarity score) is the least impactful. For `imei_unique`, high feature values are very contributive towards phone numbers being predicted as fraud and low feature values are associated with negative contribution, possibly contributing towards being predicted as non-fraudulent instead. This suggests a phone number is likely fraudulent for being linked to many imei devices, perhaps to avoid device-based blacklisting, reduce traceability and introduce noise into fraud detection models, in line with hypothesis 3.

7.4.2.2 Integrated Gradient

SHAP analysis was supplemented with IG applied directly to CARE-GNN to highlight the attribution of input features by leveraging the model's internal gradients. IG is also well-suited for deep learning models like CARE-GNN as it is able to highlight the importance of both edge and node features.

For a particular phone number (node), IG identified top features contributing towards predicting that phone number as fraud (class 1). Positive attribution value indicated the feature pushed prediction towards fraud, convincing the model the node is fraudulent while negative attribution value meant the feature pushed prediction away from fraud. Attribution value of 0 meant the feature had little or no influence on output. The magnitude of the value indicated the importance of strength with higher magnitude meaning the feature is more influential, regardless of sign.

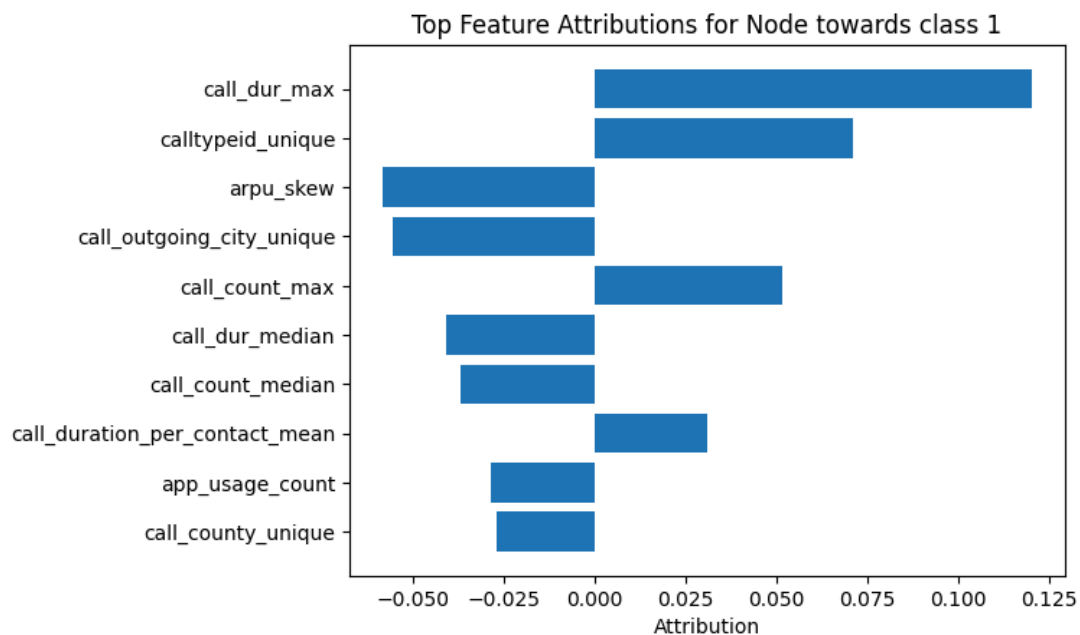


Fig 38: Feature Attributions for Node towards Fraud Class

Feature	Contribution Towards Prediction As Fraud	Observed Pattern	Insights / Linked Hypotheses
call_dur_max	Top 2 positive	Fraudulent phone numbers have higher mean maximum call duration than non-fraudulent numbers.	Associated to building trust and getting money out of victims over calls. (Channel News Asia, 2024)
calltypeid_unique		Call types include outgoing, incoming and transfer. Fraudulent numbers are involved in fewer call types.	Supported by hypothesis 4: fraudulent numbers predominantly make outgoing calls.
arpu_skewness	Top 2 negative	Features or the feature values for that particular number are not characteristic of fraud, instead indicative of non-fraud.	
call_outgoing_city_unique			

Fig 39: Observations of Feature Attributions and Insights

The absolute values (strength of importance) of attributions were averaged to identify the most important features towards fraud or non-fraud predictions across all phone numbers.

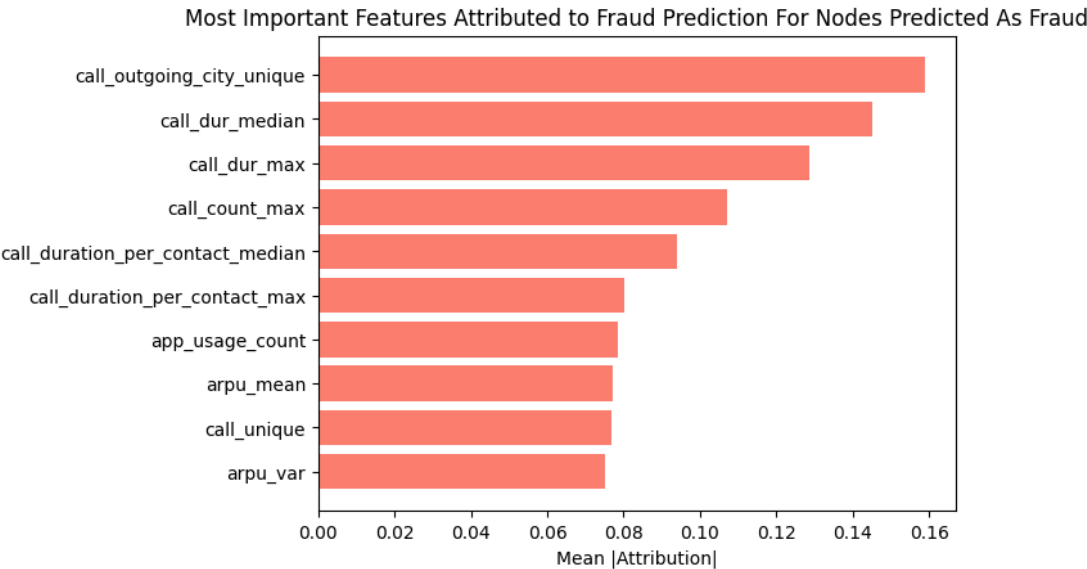


Fig 40: Feature Importance for Fraud in Prediction by absolute values of Attribution

Feature	Observed Pattern	Insights/ Linked Hypotheses
call_dur_max	May reflect deceptive tactics via phone calls	Fraud or frequent calls to saved contacts (non-fraud)
call_dur_median		
call_count_max		
app_usage_count	Fraud numbers show minimal to low app usage, primary used for making calls or messaging	Fraud numbers mainly used for calls/messages
call_unique	Fraud numbers contact many unique numbers	Supports Hypothesis 5: High variety in contacts suggests Fraud
arpu_var	Higher spending variance in fraud numbers	Supports Hypothesis 7: Fraud numbers have sporadic phone usage

Fig 41: Observations of Feature Attributions and Insights

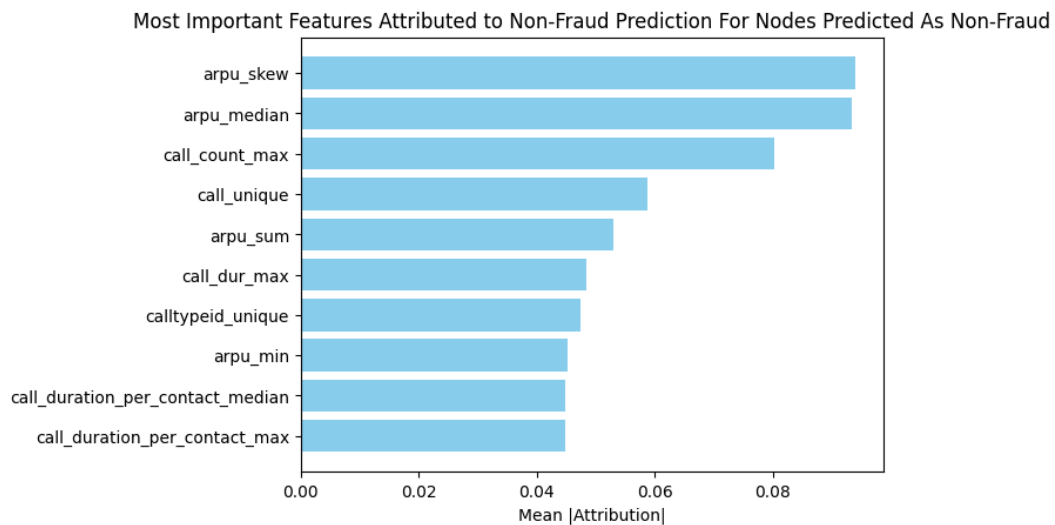


Fig 42: Feature Importance for Non-Fraud in Prediction by absolute values of Attribution

Feature	Observed Pattern	Insights/ Linked Hypotheses
call_count_max	Non-fraudulent users have more consistent contact with fewer people	Supports Hypothesis 5: Non-fraudulent users typically interact within a more limited social circle of friends, family, or colleagues, leading to fewer unique contacts
call_unique		
call_dur_max		
arpu_skew	Non-fraudulent users have more consistent ARPU	Supports Hypothesis 7: Non-fraudulent users are not like fraudulent users who exhibit sporadic phone usage

Fig 43: Observations of Feature Attributions and Insights

8. Discussion/Conclusion

8.1 Summary

We evaluated several models for fraud detection on a graph-based dataset, using RNC as our baseline. RNC provided a solid benchmark for comparison. To improve performance, we explored a feature-level ensemble approach by concatenating node embeddings generated from GCN, GAT, and GraphSAGE, and then training a RF classifier on the combined features. This ensemble model showed promising results. However, we observed that CARE-GNN outperformed all models, achieving a higher AUC score and demonstrating stronger predictive power.

This work has important implications for the telecom industry, where fraud detection is critical. By modeling user interactions—such as SMS activity, phone calls, and app usage—as graphs, and leveraging GNNs, telecom companies can more effectively identify and proactively flag potentially fraudulent behavior. This enables faster response times, reduced financial losses, and improved trust in digital communication services.

8.2 Contents

As mentioned in Section 7, our EDA was primarily supported by feature importance. However, we observed that the overall distribution of call duration features was similar between fraudulent and non-fraudulent users, suggesting that call duration alone is not a strong distinguishing factor. In Section 7.4.2.2, we found that call duration features were important for both groups, which was expected given that long call durations may still indicate fraud depending on context.

Scammers often keep victims on the line for extended periods to build psychological commitment, induce cognitive fatigue, and extract personal information for later exploitation.

Thus, while call duration features are not distinctive on their own, they remain valuable when combined with other suspicious behaviours—explaining their high ranking in our feature importance graphs.

This also applies to non-fraudulent users, where longer call durations may reflect strong personal relationships or frequent contact with close acquaintances. Therefore, these features should be seen as additive features that provide meaningful signals when used alongside other indicators.

8.3 Considerations

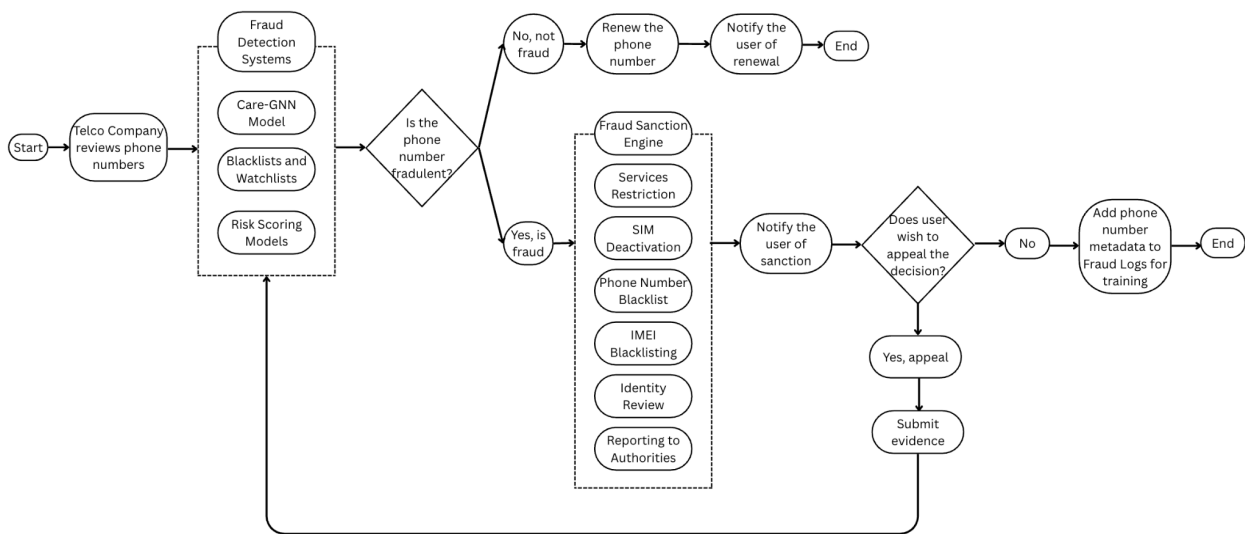


Fig 44: Pipeline for Implementation for Telecommunications Provider

For a telecommunications provider, the fraudulent number detection model could be integrated into the pipeline during periodic phone number reviews. It analyses customer registration data (ID, ARPU) and usage patterns (call/SMS behavior, app interactions), with results displayed in the fraud analytics dashboard for manual review, or an automated fraud management system for triggering actions like suspension or blacklisting fraudulent numbers.

The model detects three distinct fraud patterns: call-related, SMS-related, and app-related frauds. For call-related scams, it flags unusual calling patterns or connections to known fraud rings. In SMS-related frauds, it detects bulk messaging behavior. For app-related frauds, it identifies cross-channel activity patterns commonly associated with scams (Tay, 2023). Detection can be enhanced by integrating a rule-based system to capture extreme cases.

In the future, we encourage telecom providers to collaborate on shared blacklists of phone numbers, IMEIs, and user IDs, and enable secure customer history sharing across operators. This could strengthen fraud detection and prevention, especially for cross-border scams.

9. Appendix

9.1 Data Dictionary

Dataset	Feature Name	Description
APP (Application usage record)	phone_number_m	Subscriber's anonymized phone number (main unique identifier of user).
	app_usage_count	Number of unique apps used by the phone number over 8 months
	flow_mean	Average data usage across all app interactions over 8 months for the phone number. (in MB)
	flow_median	Median data usage across all app interactions over 8 months for the phone number. (in MB)
	flow_min	Minimum data usage across all app interactions over 8 months for the phone number. (in MB)
	flow_max	Maximum data usage across all app interactions over 8 months for the phone number. (in MB)
	flow_var	Variance in data usage across all app interactions over 8 months for the phone number. (in MB)
	flow_sum	Total data usage across all app interactions over 8 months for the phone number. (in MB)
	flow_month	Average data usage per month for the phone number (flow_sum / months_count)
	months_count	Number of months the user was active (from August 2019 to March 2020)

SMS (Short Message Service subset)	sms_count	Total number of SMS records per phone number across 8 months (messages sent and received).
	sms_nunique_contact	Number of unique contacts the subscriber exchanged SMS with across 8 months.
	sms_date_nunique	Total number of unique dates an SMS was sent over 8 months
	sms_hour_mode	Most frequent hour (mode) SMS messages were sent.
	sms_hour_mode_count	Count of SMS messages sent during the most frequent hour.
	sms_hour_nunique	Number of unique hours SMS messages were sent.
	sms_day_mode	Most frequent day of the month (1st-31st) on which SMS messages were sent. E.g. The most frequent day is the 1st of each month.
	sms_day_mode_count	Count of SMS messages sent on the most frequent day of the month.
	sms_day_nunique	Number of unique days of the month on which SMS messages were sent.
	sms_dayname_mode_count	Most frequent day of the week (mode) on which SMS messages were sent. (Monday-Sunday)
	sms_dayname_nunique	Number of unique days of the week on which SMS messages were sent. (Monday-Sunday)
	sms_rate	Average number of messages per unique contact across 8 months (sms_count/sms_nunique_contact)
	sms_contacts_proportion	Proportion of unique contacts relative

		to the total SMS messages sent. Close to 1 means interaction with many different contacts. Close to 0 means interactions with the same few contacts.
	sms_calltype1_count	Count of outgoing SMS messages sent (calltype_id = 1) across 8 months.
	sms_calltype2_count	Count of incoming SMS messages received (calltype_id = 2) across 8 months.
	sms_calltype2_proportion	Proportion of outgoing SMS messages sent (calltype_id = 2) to total SMS messages across 8 months.
VOC (Voice Record)	imei_count	Number of devices that the phone number was used on across 8 months. International Mobile Equipment Identity (IMEI) is a unique identifier for the device used by the subscriber.
	imei_list	A list of unique IMEI numbers associated with the phone number.
	call_count	Total number of voice call records across 8 months (incoming + outgoing calls).
	call_unique	Number of unique phone numbers the subscriber has called or received calls from across 8 months.
	call_city_unique	Number of unique cities the phone number has made or accepted calls across 8 months.
	call_county_unique	Number of unique counties the phone number has made/accepted calls across 8 months.

	call_count_mean	Average number of calls made to each unique contact across 8 months. (so group by phone_no_m, then group by opposite_no_m, then get the count of each pair, then find mean of all the counts within that phone_no_m)
	call_count_median	Median number of calls made to each unique contact across 8 months
	call_count_max	Maximum number of calls made to a single unique contact (opposite phone number) across 8 months.
	call_dur_mean	Average duration of calls across 8 months (in seconds).
	call_dur_median	Median duration of calls across 8 months (in seconds).
	call_dur_max	Maximum duration of a call across 8 months (in seconds).
	call_dur_min	Minimum duration of a call across 8 months (in seconds).
	voc_hour_mode	Most frequent hour for calls across 8 months.
	voc_hour_mode_count	Call count during most frequent hour (mode).
	voc_hour_nunique	Number of unique hours the subscriber makes/receives calls.
	voc_day_mode	Most frequent day of the month for calls (1st-31st).
	voc_day_mode_count	Call count on most frequent day (mode).
	voc_date_unique	Number of unique days the subscriber makes/receives calls over 8 months.
	voc_dayname_mode_count	Count of calls made on the most frequent day of the week (mode).

		(Monday-Sunday)
	voc_dayname_unique	Number of unique days of the week the subscriber makes/receives calls across 8 months.
	calltypeid_unique	Number of call types used by the subscriber. Call types: - Outgoing (person is the caller) - Incoming (person is the receiver) - Transfer 1 means one type only, 2 means 2 types, 3 means all 3 types.
	voc_calltype1_count	Count of outgoing calls (calltype_id = 1) across 8 months.
	voc_calltype2_count	Count of incoming calls (calltype_id = 2) across 8 months.
	voc_calltype3_count	Count of transfer calls (calltype_id = 3) across 8 months.
	voc_calltype1_proportion	Proportion of outgoing calls made (calltype_id = 1) to total calls across 8 months.
	call_outgoing_city_unique	Number of unique cities the phone number made outgoing calls to (calltype_id = 1) across 8 months.
	call_outgoing_county_unique	Number of unique counties the phone number made outgoing calls to (calls with calltype_id = 1) across 8 months.
	call_duration_per_contact_median	Median total call duration of calls made to each unique contact across 8 months.
	call_duration_per_contact_max	Maximum total call duration of calls made to each unique contact across 8 months.

		months.
	call_duration_per_contact_mean	Mean total call duration per unique contact across 8 months, calculated as the product of mean call duration across 8 months and call count divided by the unique contacts.
USER (Consumption record)	arpu_mean	Average monthly spending (ARPU) in dollars over 8 months. ARPU (Average Revenue Per User) typically reflects the subscriber's spending behavior on telecom services.
	arpu_var	Variance in monthly spending.
	arpu_max	Max monthly spending.
	arpu_min	Min monthly spending.
	arpu_median	Median monthly spending.
	arpu_sum	Total spending over 8 months.
	arpu_skew	Skewness of spending distribution (high skew might indicate irregular behaviour).
	arpu_sem	Standard error of the mean (Spread of spending data).
	idcard_count	Number of phone numbers registered under the same ID.
	label	Fraudulent number label: 1 for fraudulent 0 for non-fraudulent

9.2 Additional Graph Construction

Type of Graph	Description
Cosine Similarity (Main)	Refer to Graph Construction
Multi-Relation Communication Graph	A heterogeneous graph to model relationships between phone numbers based on similarity, shared IMEIs, and call and sms interactions. Node Features & Labels: - Extract numerical and categorical features from the dataset, normalizing numerical attributes and one-hot encoding categorical ones. - Labels are aligned with node indices for supervised learning. Graph Relations: Similarity Graph: Constructed using cosine similarity between feature vectors, selecting 14 most similar and 6 least similar neighbors per node. Shared IMEI Graph: Nodes sharing IMEIs are connected, forming edges based on common device usage. Call Interaction Graph: Nodes are linked if a phone number appears in another's call records, capturing direct communication patterns. Sparse Representation: - Adjacency matrices are converted into sparse representations for efficient storage. - Graph structures are saved as adjacency lists, alongside node features and labels for downstream processing.

	○ This graph structure enables learning node embeddings that integrate behavioral, device, and interaction-based signals for fraud detection.
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9.3 All Model Results

1. Models tried but not used

Alg	Graph Method	Metrics	Normal	Fraud
GCN embeddings + RF	Cosine Similarity	Pr	0.94	0.87
		Rc	0.90	0.78
		F1	0.92	0.82
		AUC-ROC	0.9377	
		Accuracy	0.89	
GAT embeddings + RF	Cosine Similarity	Pr	0.95	0.88
		Rc	0.90	0.78
		F1	0.92	0.83
		AUC-ROC	0.9329	
		Accuracy	0.89	
GraphSage embeddings + RF	Cosine Similarity	Pr	0.96	0.91
		Rc	0.93	0.84
		F1	0.94	0.88
		AUC-ROC	0.9295	
		Accuracy	0.92	

2. Baseline Model

Alg	Graph Method	Metrics	Normal	Fraud	Overall (weighted avg)
Relational Neighbour Classifier	Cosine similarity	Pr	0.87	0.92	0.88
		Rc	0.97	0.70	0.88
		F1	0.91	0.79	0.87
		AUC-ROC	0.8321		
		Accuracy	0.88		

3. Challenger Models

Alg	Graph Method	Metrics	Normal	Fraud	Overall (weighted avg)
Care-GNN	Relational Graph	Pr	0.93	0.87	0.91
		Rc	0.94	0.85	0.91
		F1	0.93	0.86	0.91
		AUC-ROC	0.9502		
		Accuracy	0.91		
Feature-Level Ensemble – concatenation (node embeddings from GCN, Graphsage, GAT) (embeddings to train Random Forest)	Cosine similarity	Pr	0.97	0.94	
		Rc	0.93	0.84	
		F1	0.95	0.89	
		AUC-ROC	0.9420		
		Accuracy	0.93		

4. Fine Tuning of Potential Final Model

Care-GNN

Alg	Graph Method	Metrics	Normal	Fraud	Overall (weighted avg)
Care-GNN (Fine-tuned)	Relational Graph	Pr	0.94	0.87	0.91
		Rc	0.94	0.86	0.91
		F1	0.94	0.87	0.91
		AUC-ROC	0.9534		
		Accuracy	0.91		

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